AI, Ethics & Professional Issues

From Code of Conduct to AI Alignment

Part 1: Foundations

Ethics, Liability, and the History of Software Failure

Trade vs. Profession

Why are we here?

Computing began as a craft, but has evolved into a profession, similar to medicine or law.

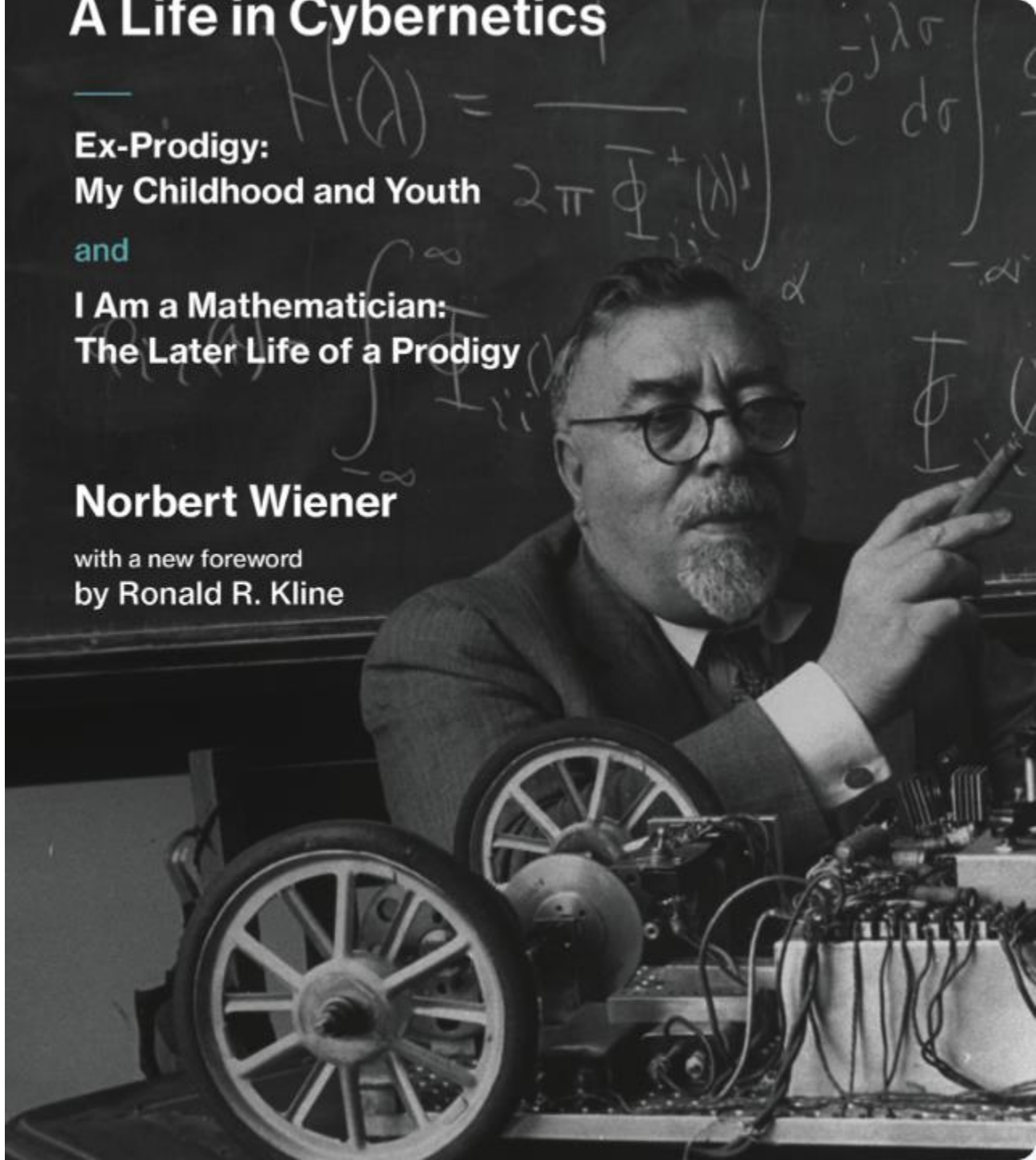
- **Specialized Education:** Requires formal training, not just apprenticeship.
- **Public Duty:** The primary obligation is to public safety, not just the employer's profit.
- **Code of Ethics:** A shared standard of conduct enforced by peers.

Norbert Wiener (1948)

The Prophet of Cybernetics

Decades before the internet, Wiener predicted the "Second Industrial Revolution" of automation.

- **Prediction:** Machines would devalue human brain labor just as steam engines devalued muscle labor.
- **Warning:** "We must use our machines for human ends, not let them use us." He foresaw the ethical crisis of automation and unemployment.



The ELIZA Effect (1966)

The Illusion of Understanding

Joseph Weizenbaum created a simple chatbot that mimicked a therapist. To his shock, users (including his secretary) formed deep emotional bonds with it.

- **The Lesson:** Humans are hard-wired to project intelligence and empathy onto machines, even when none exists.
- **Ethics:** Is it ethical to create software that deceives users into thinking it "cares"?

Context: The Architecture of Deception

Pattern Matching

ELIZA scanned for keywords (e.g., "Mother", "Family"). If none were found, it gave a generic response ("Tell me more").

Substitution

It swapped pronouns to reflect the user's input back at them. "I am sad" became "Why are you sad?".

The Ethical Breakpoint

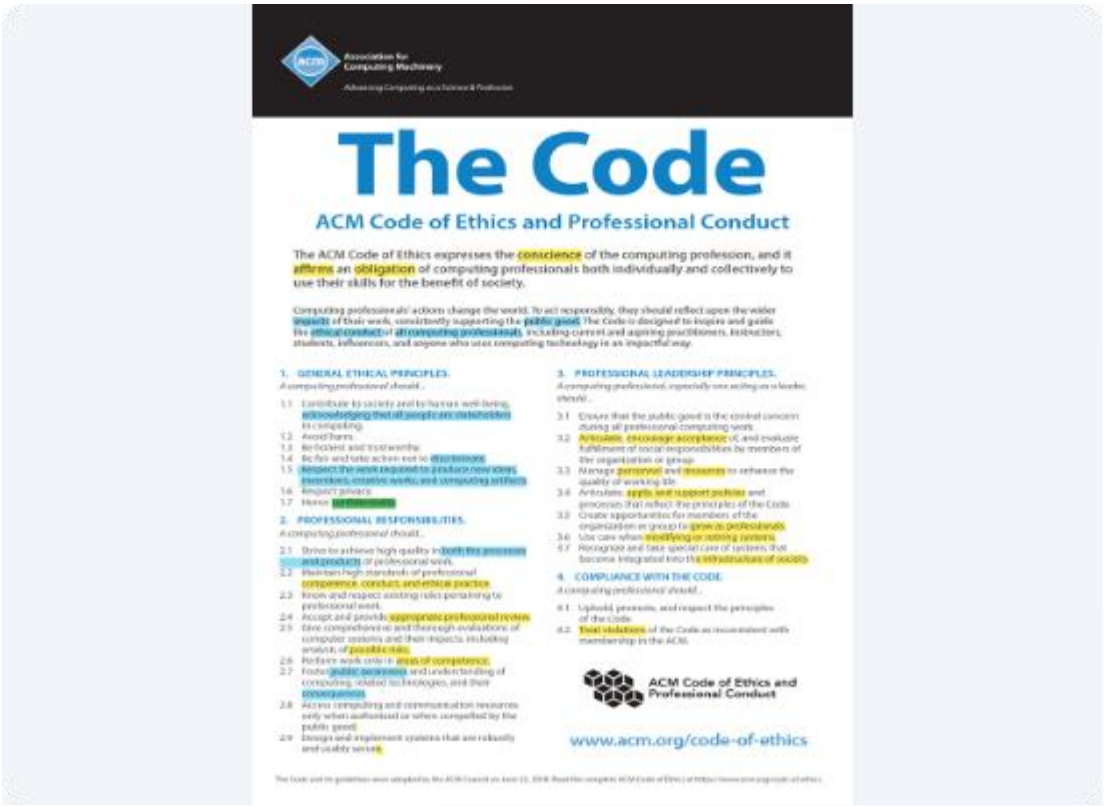
The code was simple (200 lines). The ethical issue wasn't the code, but the **emotional vulnerability** of the user vs. the developer's intent.

The Foundation: Codes of Ethics

More than just rules

As software "eats the world," developers hold immense power. Professional bodies like ACM and IEEE have established codes to guide this responsibility.

- **Public Interest:** The safety and welfare of the public is paramount.
- **Competence:** Only undertake work you are qualified to perform.
- **Harm Avoidance:** Minimize negative consequences of systems (data loss, privacy violation).



Association for Computing Machinery

Advancing Computing as a Science & Profession

The Code

ACM Code of Ethics and Professional Conduct

The ACM Code of Ethics expresses the conscience of the computing profession, and it affirms an obligation of computing professionals both individually and collectively to use their skills for the benefit of society.

Computing professionals' actions change the world. To act responsibly, they should reflect upon the wider impacts of their work, consistently supporting the public good. The Code is designed to inspire and guide the ethical conduct of all computing professionals, including current and aspiring practitioners, instructors, students, influencers, and anyone who uses computing technology in an impactful way.

1. GENERAL ETHICAL PRINCIPLES.

A computing professional should...

1.1 Contribute to society and to human well-being, acknowledging that all people are stakeholders in computing.

1.2 Avoid harm.

1.3 Be honest and trustworthy.

1.4 Be fair and take action not to discriminate.

1.5 Respect the work required to produce new ideas, inventions, creative works, and computing artifacts.

1.6 Respect privacy.

1.7 Honor confidentiality.

2. PROFESSIONAL RESPONSIBILITIES.

A computing professional should...

2.1 Strive to achieve high quality in both the processes and products of professional work.

2.2 Maintain high standards of professional competence, conduct, and ethical practice.

2.3 Know and respect existing rules pertaining to professional work.

2.4 Accept and provide appropriate professional review.

2.5 Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks.

2.6 Perform work only in areas of competence.

2.7 Foster public awareness and understanding of computing, related technologies, and their consequences.

2.8 Access computing and communication resources only when authorized or when compelled by the public good.

2.9 Design and implement systems that are robustly and usably secure.

3. PROFESSIONAL LEADERSHIP PRINCIPLES.

A computing professional, especially one acting as a leader, should...

3.1 Ensure that the public good is the central concern during all professional computing work.

3.2 Articulate, encourage acceptance of, and evaluate fulfillment of social responsibilities by members of the organization or group.

3.3 Manage personnel and resources to enhance the quality of working life.

3.4 Articulate, apply, and support policies and processes that reflect the principles of the Code.

3.5 Create opportunities for members of the organization or group to grow as professionals.

3.6 Use care when modifying or retiring systems.

3.7 Recognize and take special care of systems that become integrated into the infrastructure of society.

4. COMPLIANCE WITH THE CODE.

A computing professional should...

4.1 Uphold, promote, and respect the principles of the Code.

4.2 Treat violations of the Code as inconsistent with membership in the ACM.

ACM Code of Ethics and Professional Conduct

www.acm.org/code-of-ethics

Theory: How to Decide?

The Trolley Problem in Code

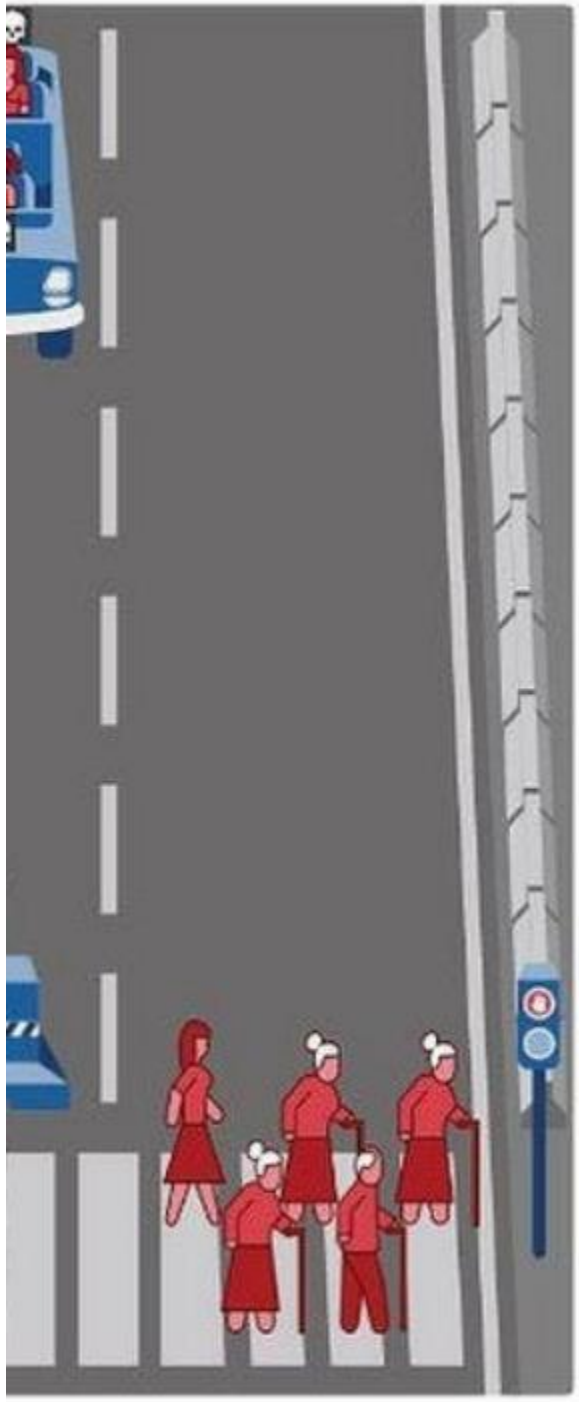
Utilitarianism: Maximize total happiness/minimize total harm.

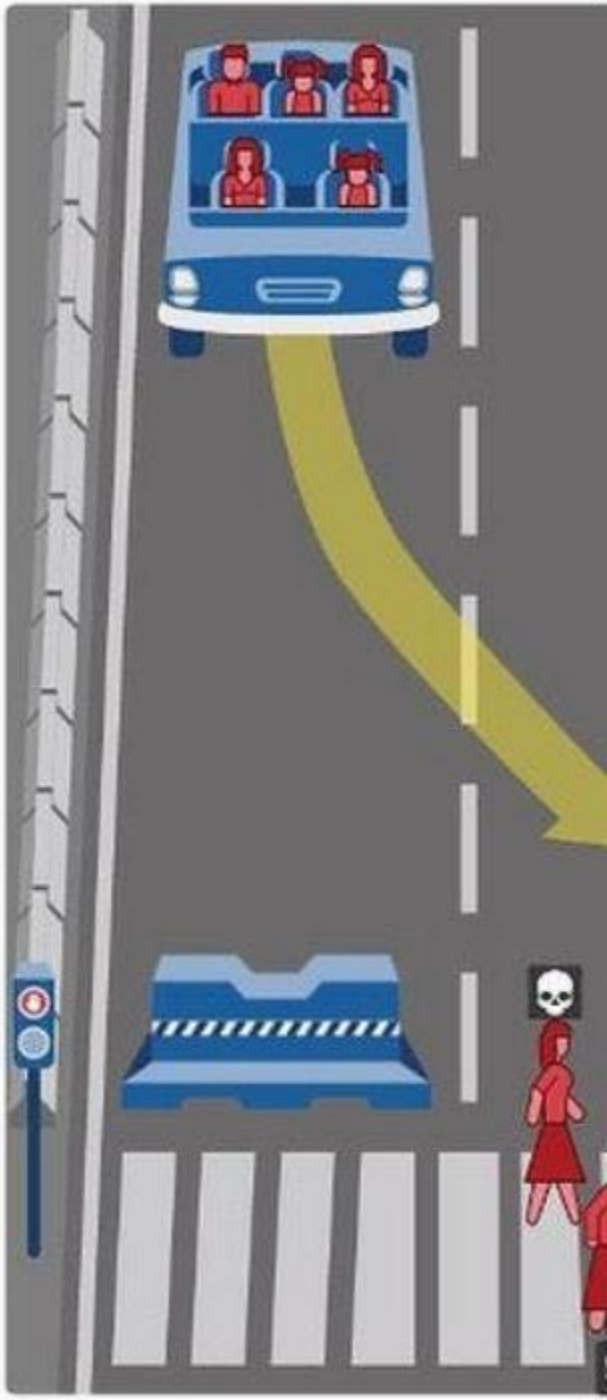
Decision: Swerve to kill 1 and save 5.

Deontology: Duty-based ethics. Certain actions are inherently right or wrong, regardless of consequences.

Decision: Never intentionally kill innocent people, even to save others.

Relevance: How do we program "morality" into an autonomous vehicle?





The Open Source Ethos

Free as in Freedom

Richard Stallman launched the Free Software movement not because he hated money, but because he believed proprietary code violated a user's moral right to repair and understand their tools.

- **The Printer Story:** Stallman just wanted to fix a paper jam, but the closed source code prevented him.
- **Impact:** Created the concept of "Copyleft" (GPL), forcing code improvements to remain public commons.

Accessibility (a11y)

Ethics of Exclusion

If you build a digital service that is critical for life (banking, government) but inaccessible to the blind, you have effectively revoked their citizenship.

- **WCAG:** Web Content Accessibility Guidelines are not just "nice to have," they are often a legal requirement.
- **Universal Design:** Designing for disabilities often improves the product for everyone (e.g., subtitles, high contrast).

The Digital Divide

Inequality of Access

Technology amplifies power. Those without access to high-speed internet or modern hardware fall further behind in education and economy.

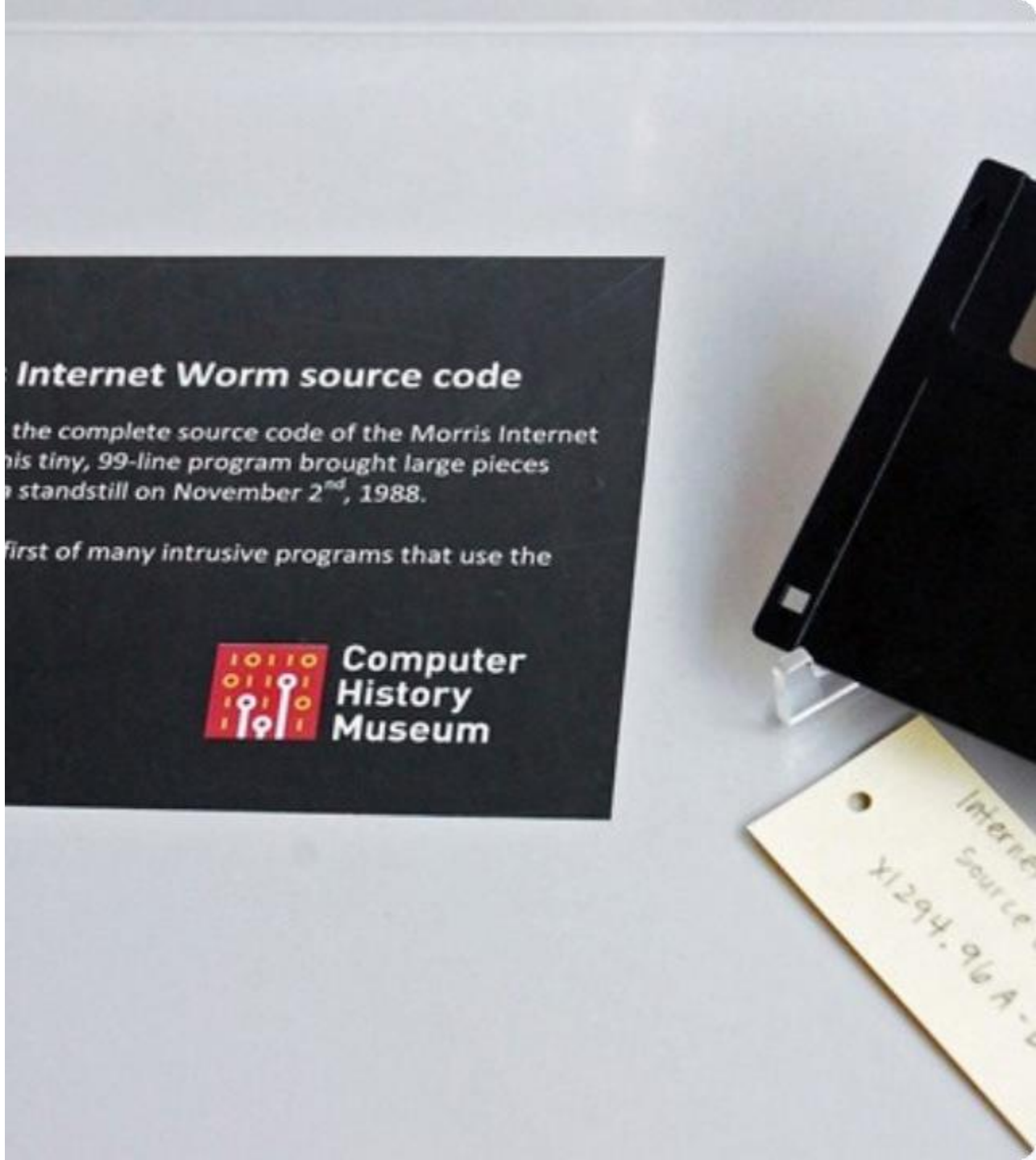
- **Global:** The gap between developed and developing nations.
- **Local:** The "Homework Gap" where students sit in McDonald's parking lots for WiFi.
- **Ethical Duty:** Should internet access be a human right?

Case Study: The Morris Worm (1988)

The First Felony

Robert Tappan Morris wrote a program to gauge the size of the internet. Due to a coding error, it replicated uncontrollably, crashing 10% of the entire internet.

- **Intent vs. Outcome:** He didn't mean to destroy, but his negligence caused massive damage.
- **Precedent:** He became the first person convicted under the Computer Fraud and Abuse Act, setting the stage for cybercrime law.



Context: The "gets()" Vulnerability

The Technical Failure

The worm exploited the fingerd service using the C function gets(), which does not check buffer length.

- **The Exploit:** By sending more data than the buffer could hold, Morris overwrote the return address on the stack (Buffer Overflow).
- **The Logic Error:** The worm was supposed to check if a machine was already infected. To avoid false positives, Morris made it re-infect 1 out of 7 times regardless. This probabilistic rate was too high, creating a denial-of-service.

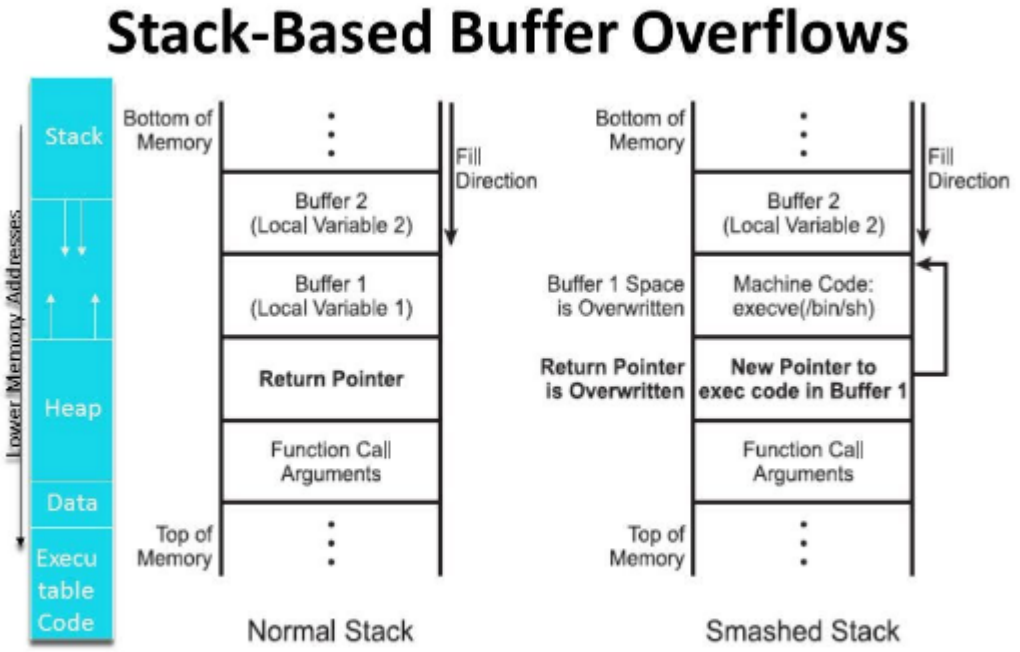


EXHIBIT 10.2 A normal stack and a stack with a buffer overflow.

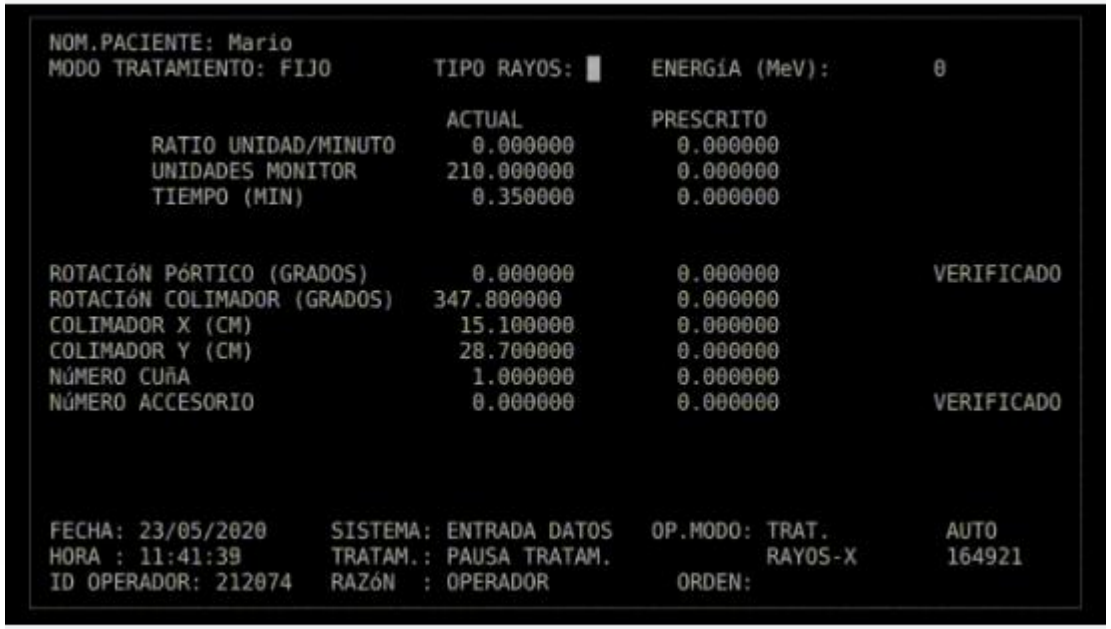
Case Study: Therac-25

When Bugs Kill

Between 1985 and 1987, a radiation therapy machine massively overdosed six patients, leading to deaths. It is the defining case study for software safety.

- **The Error:** A race condition in the code that occurred when operators typed too fast.
- **The Failure:** The software relied on a counter variable that overflowed.
- **Lesson:** "Software is not just a tool; it is a critical infrastructure."

Context: Removing the Safety Net



The Ethical Breakpoint

The previous model (Therac-20) had the *same software bugs*, but hardware interlocks prevented the beam from firing. The Therac-25 removed these physical locks to save cost, relying entirely on software.

- **UI Design:** The cryptic error "Malfunction 54" gave operators no indication of a lethal overdose, causing them to repeat the dose.
- **Lesson:** Never remove hardware safety based on "perfect" software assumptions.

The $370 Million Integer Overflow

Ariane 5 Flight 501 (1996)

37 seconds after launch, the rocket flipped 90 degrees and disintegrated. The cause? A piece of code reused from the older Ariane 4.

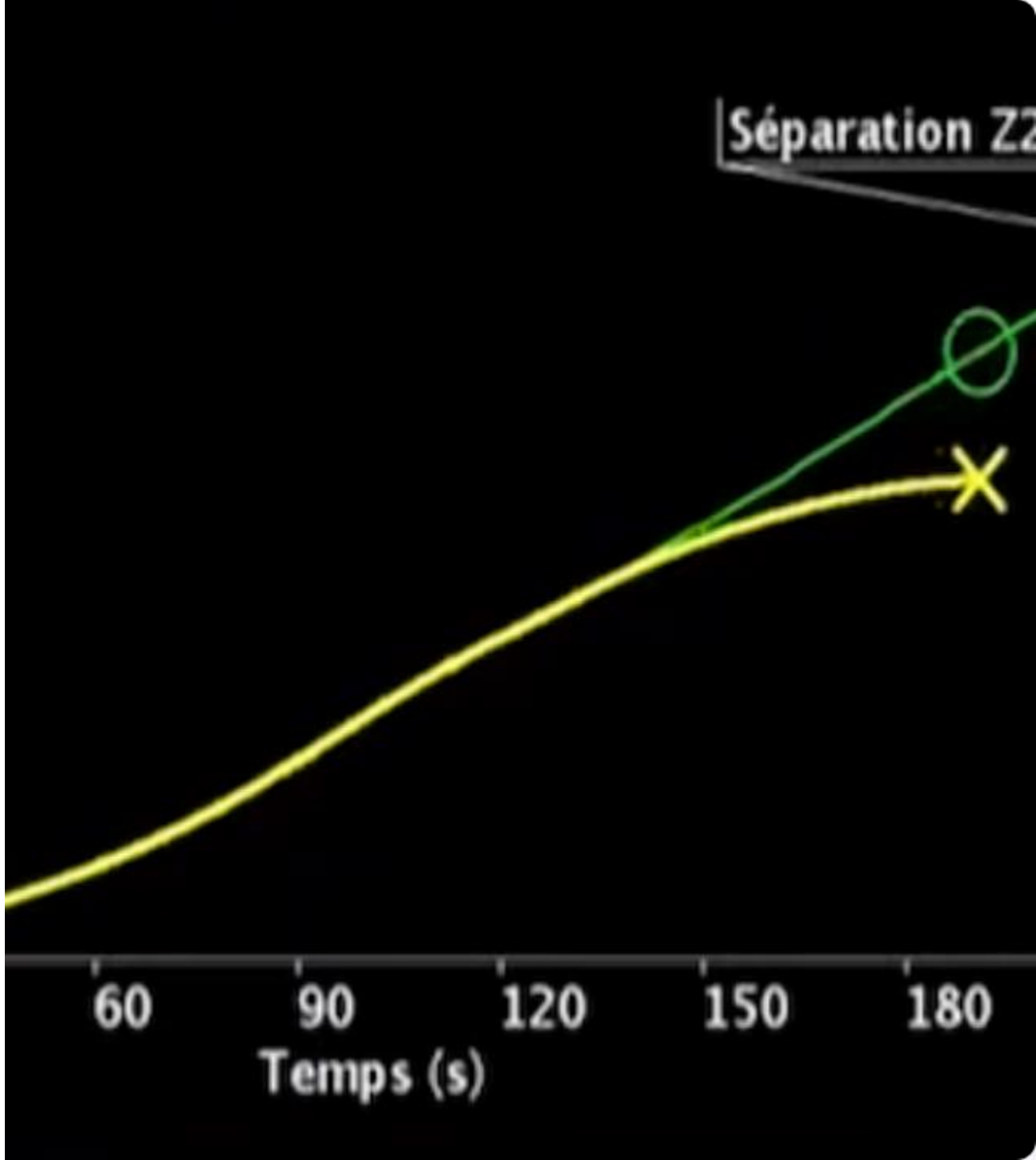
- **Technical Fault:** Attempting to stuff a 64-bit float velocity value into a 16-bit signed integer.
- **Professional Issue:** Inadequate testing of legacy code in a new environment. Blindly copying code without context is professional negligence.

Context: Dangerous Assumptions

Why did it fail?

The code monitored "horizontal bias" (velocity). The Ariane 5 was much faster than the Ariane 4. The velocity value physically exceeded the maximum value of a 16-bit integer (32,767).

- **The Exception:** The overflow triggered an exception, which dumped debug data into the guidance channel. The steering engines interpreted this debug data as a command to turn 90 degrees.
- **Ethical Takeaway:** Blindly reusing code without re-verifying its boundary conditions is negligence.



When the Code is the Crime

Volkswagen "Dieselgate"

Engineers wrote software specifically designed to detect when the car was being tested (e.g., steering wheel not moving, consistent speed) and switch the engine to a low-emission mode.

The Ethical Breach

This was not a bug. It was a feature requested by management and implemented by developers. It highlights the conflict between following orders and professional integrity.

Result: Billions in fines and prison sentences for executives.

Context: The Defeat Device

How it worked

The Engine Control Unit (ECU) monitored sensors for a very specific "Test Cycle" profile:

- **Triggers:** Hood open + Linear speed (rollers) + Steering wheel angle = 0.
- **Action:** If conditions met, activate "Dyno Calibration" mode (rich fuel mix, high AdBlue usage) to pass the test.
- **Reality:** On the road (steering moving), it switched to "dirty" mode for better mileage but 40x legal NOx emissions.

Dark Patterns: Weaponized UX

Designing for Deception

Dark patterns are UI designs crafted to trick users into doing things they didn't mean to, like buying insurance or sharing contacts.

- **Roach Motel:** Easy to get in (subscribe), impossible to get out (cancel).
- **Confirmshaming:** "No, I don't want to save money."
- **Privacy Zuckering:** Tricking users into sharing more info than intended.

Surveillance Capitalism

You are the Product

Shoshana Zuboff's concept describes a new economic order that claims human experience as free raw material for hidden commercial practices of extraction, prediction, and sales.

- **The Goal:** Not just to predict your behavior, but to modify it to suit advertisers.
- **The Cost:** The loss of human agency and the right to a private future.

Whistleblowing & Loyalty

Edward Snowden & PRISM

In 2013, a contractor revealed global mass surveillance programs run by the NSA. It fundamentally changed the conversation on digital privacy.

- **The Dilemma:** He signed an NDA (legal obligation) but witnessed what he believed were constitutional violations (moral obligation).
- **Impact:** It forced tech giants to encrypt data by default, moving the industry from "trust us" to "verify us."

Discussion: The "Launch or Delay" Dilemma

The Scenario

You are a lead developer. Your company is launching a major product tomorrow. Marketing has spent $500k on the launch event.

At 8:00 PM, you find a security vulnerability. It affects only 0.1% of users but could potentially leak personal data. Fixing it requires a full 3-day delay.

1. Do you delay the launch?

2. Do you patch it silently later?

3. If management forces the launch, what do you do?

Part 2: The AI Paradigm

From Deterministic Logic to Probabilistic Models

The AI Revolution in Development

From Explicit to Probabilistic

Traditional programming relies on explicit logic (If/Else), where the developer defines every rule. AI development shifts this to a probabilistic model (Data → Model → Prediction).

The Core Shift:

- **Old:** Logic correctness is the primary ethical check.
- **New:** Data integrity, model fairness, and societal impact are the new professional standards.

Ethics in the AI Production Lifecycle

1. Scoping

Techno-solutionism: Is AI the right tool? Should this problem be solved by an algorithm at all?

2. Data Collection

The Foundation: Issues of copyright, user consent (GDPR), and representational bias occur here.

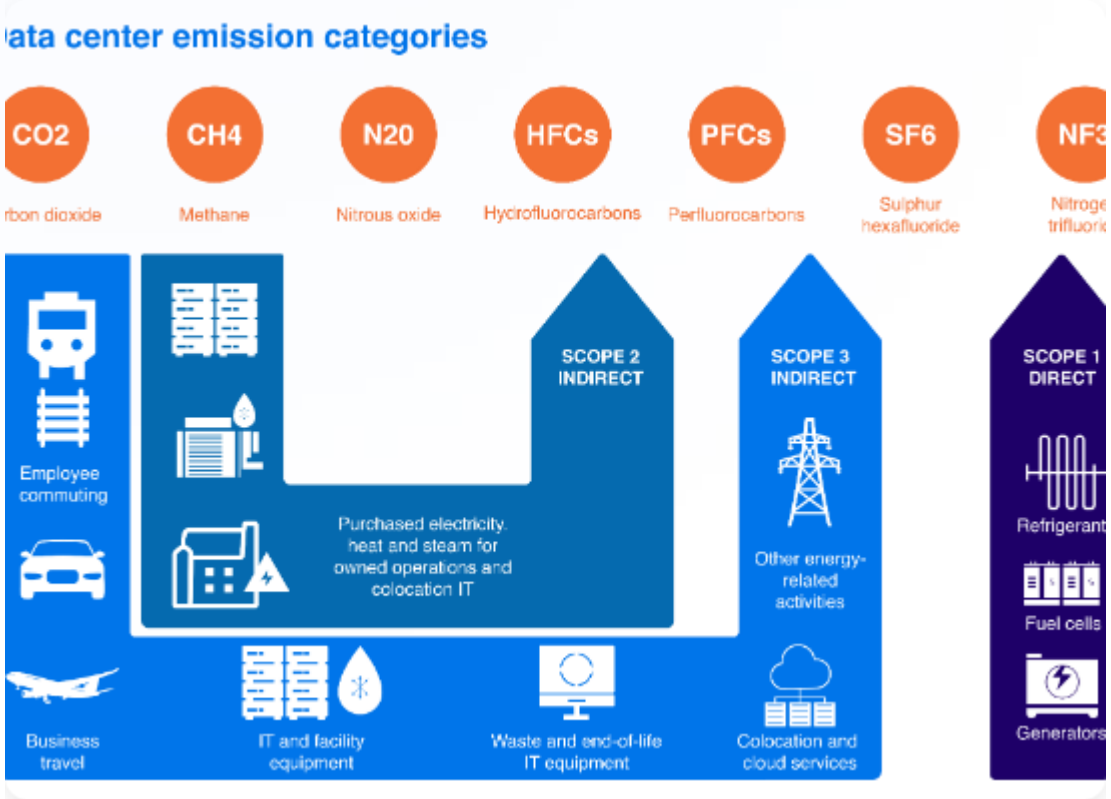
3. Training

The Black Box: Environmental costs of computing and lack of interpretability in the resulting model.

4. Deployment

Feedback Loops: Automation bias, misuse by bad actors, and long-term societal impact.

Hidden Costs: Labor & Carbon



The "Ghost Work" Behind AI

AI isn't magic; it's supervised. Reinforcement Learning (RLHF) relies on low-paid workers in developing nations to label toxic content, often causing psychological trauma.

Environmental Impact

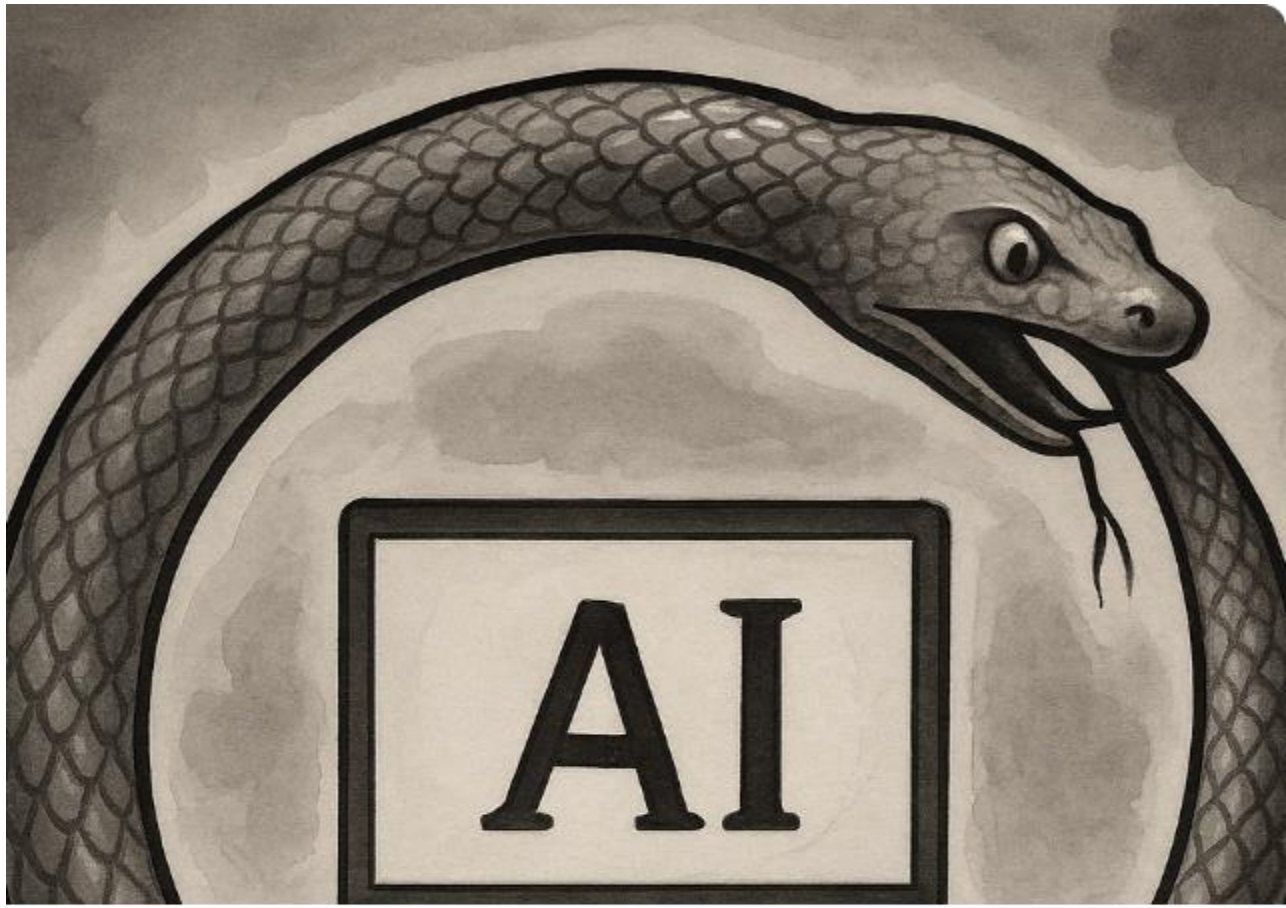
Training a single large transformer model can emit as much carbon as five cars do in their entire lifetimes. Efficiency is now an ethical metric.

Theory: Model Collapse

The Ouroboros of Data

Generative AI models are trained on the open web. As the web becomes flooded with AI-generated content, future models will be trained on the output of past models.

- **The Risk:** "Garbage in, garbage out" becomes an amplifying feedback loop. Models lose variance, creativity, and eventually degenerate into nonsense.
- **Practice:** The need for "provenance" mechanisms to distinguish human vs. synthetic data.



The Training Data Dilemma

Modern AI models (LLMs) are fed petabytes of internet data. This creates massive friction with existing data protection laws like GDPR.

- **Consent:** Did the users agree to have their blog posts or photos train a commercial model?
- **PII Leakage:** Models can inadvertently memorize and regurgitate private info (names, addresses).
- **The "Right to be Forgotten":** Deleting a database row is easy. Making a neural network "unlearn" a specific concept is mathematically difficult.

Law: GDPR Article 22

Automated Individual Decision-Making

The GDPR grants EU citizens specific protections against "Black Box" algorithms.

> "The data subject shall have the right not to be subject to a decision based solely on automated processing..."

- **Right to Explanation:** You must be able to explain *how* the decision was reached.
- **Human Intervention:** A user can demand a human review the AI's decision.

Discussion: The "Right to be Forgotten"

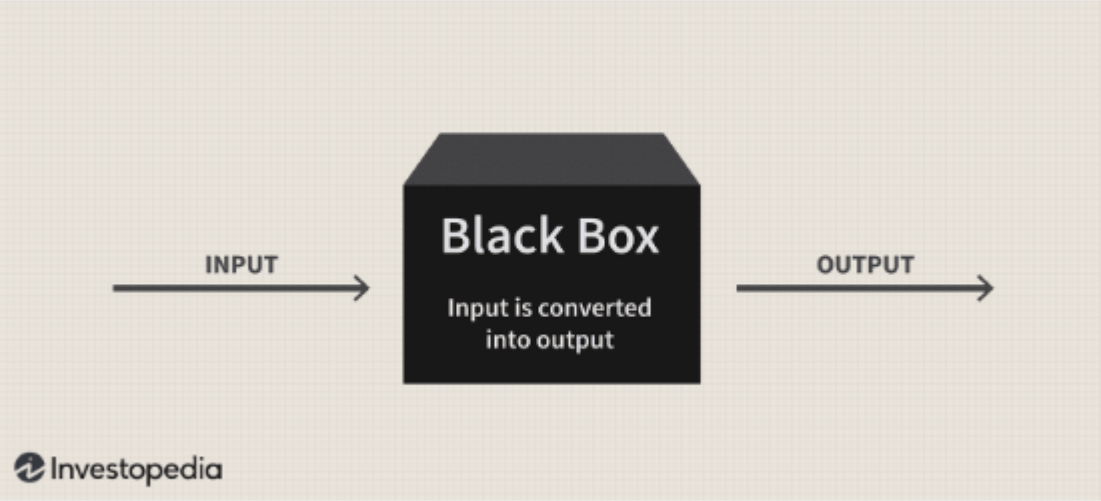
In a traditional database, we can run DELETE * FROM users WHERE id=123. In a neural network, data is stored as weighted connections, not rows.

Question for Class

If an LLM has "memorized" your private data, is it technically possible to delete it without retraining the entire model from scratch (costing millions)?

(Hint: Look up "Machine Unlearning")

The "Black Box" Problem



Opacity vs. Trust

In traditional software, we can trace a bug to a specific line of code. In Deep Learning, decisions are made by billions of floating-point parameters.

- **Explainable AI (XAI):** A growing field focused on making AI decisions interpretable.
- **Why it matters:** If an AI denies a loan or rejects a job application, professional ethics (and law) demand we explain *why*.

Algorithmic Bias & Discrimination

Selection Bias

When the training data is not representative of the real world. (e.g., Face recognition trained mostly on light-skinned men fails on other demographics).

Historical Bias

Data accurately reflects the world, but the world itself is biased. (e.g., Hiring data showing men in leadership roles leads AI to de-prioritize women).

Automation Bias

The human tendency to over-trust the machine. "The computer said so, so it must be true," leading to a lack of critical human oversight.

Discussion: Bias in Justice

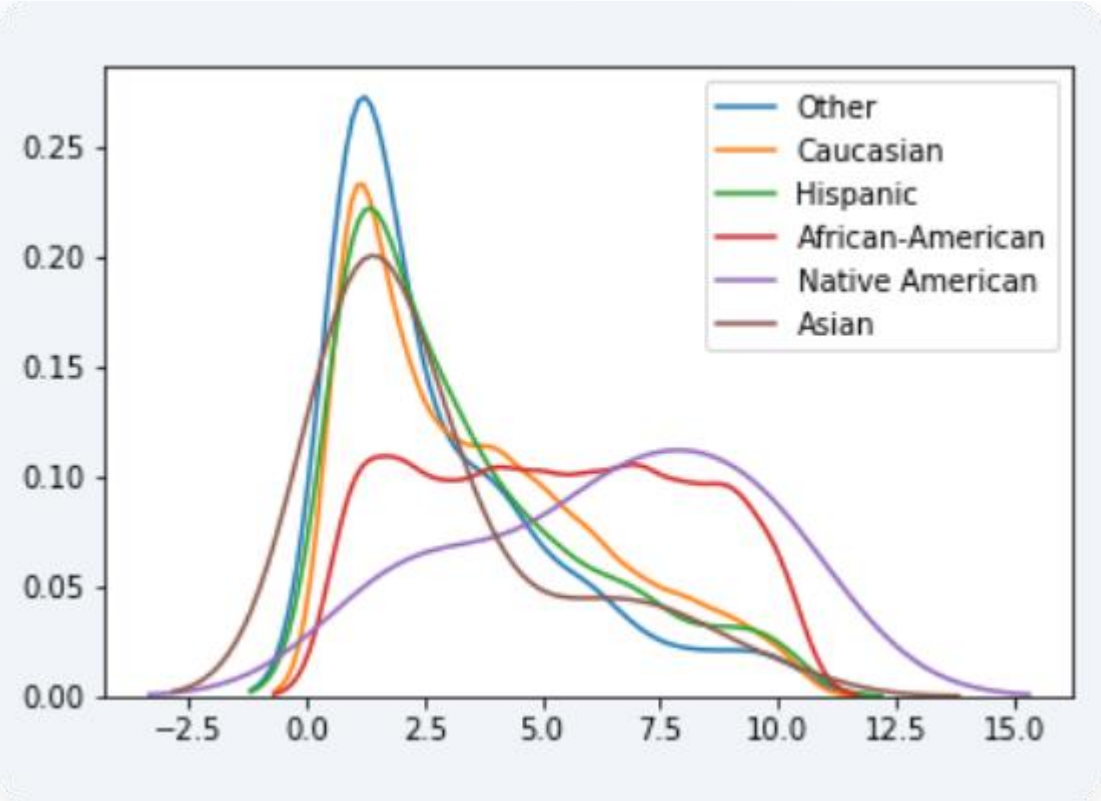
The COMPAS Algorithm

Used in US courts to predict recidivism (likelihood of re-offending). ProPublica analysis found:

- Black defendants were roughly **2x more likely** to be wrongly labeled as "High Risk" compared to White defendants.
- The algorithm didn't use race as a variable, but used proxy variables (zip code, friends' arrest records).

Is "math" neutral if the history it learns from is biased?

Context: Understanding False Positives



The Statistical Failure

The core ethical issue wasn't that the algorithm was "racist" by design, but that it optimized for the wrong metric.

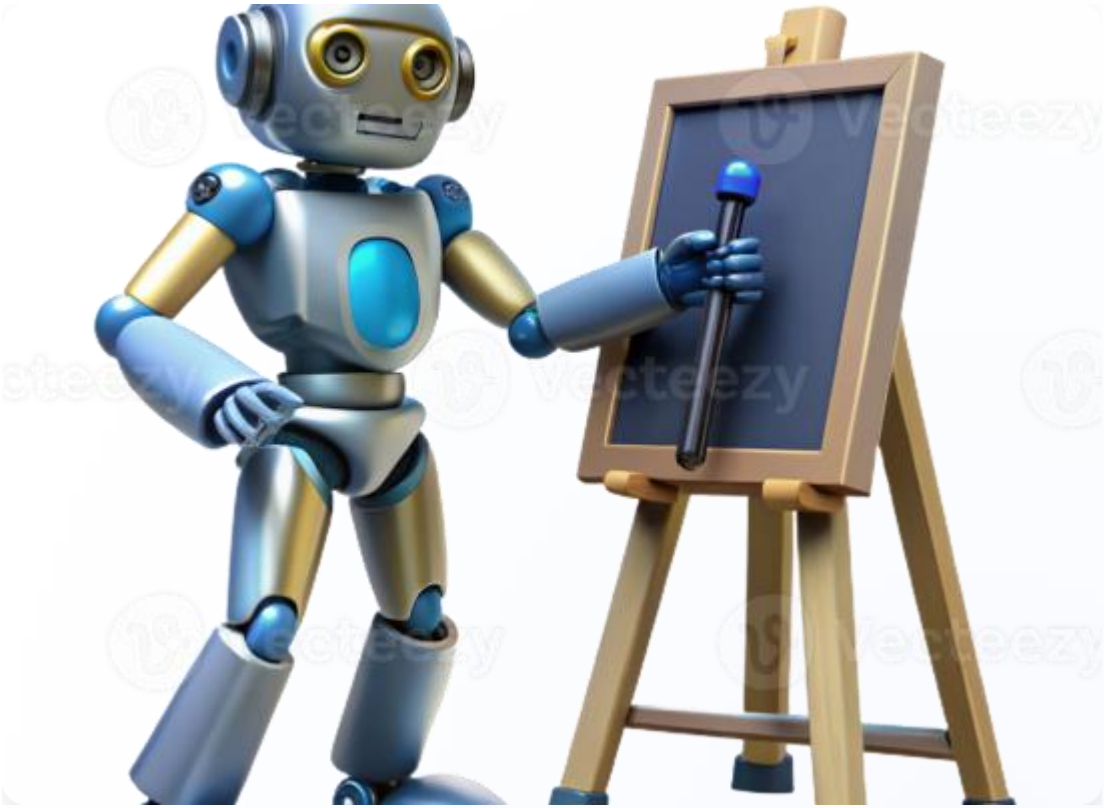
- **False Positives:** High risk score, but did NOT re-offend. Rate for African Americans: 44.9%. Rate for Whites: 23.5%.
- **Impact:** Innocent people were denied parole based on a statistical error that disproportionately affected one group.

Intellectual Property in GenAI

Who Owns the Code?

Tools like GitHub Copilot are trained on open-source repositories. This raises critical legal questions:

- **License Laundering:** Does the AI strip GPL/MIT licenses from the code it generates?
- **Ownership:** Current rulings (e.g., Thaler v. Perlmutter) suggest AI-generated works cannot be copyrighted because they lack "human authorship."
- **Impact:** Developers must be wary of pasting proprietary code into public AI tools.



Law: Fair Use vs. Theft

The Argument for "Fair Use"

AI companies argue that training models is like a student reading a book. They are learning patterns, not copying text. This is "transformative" use.

The Argument for "Theft"

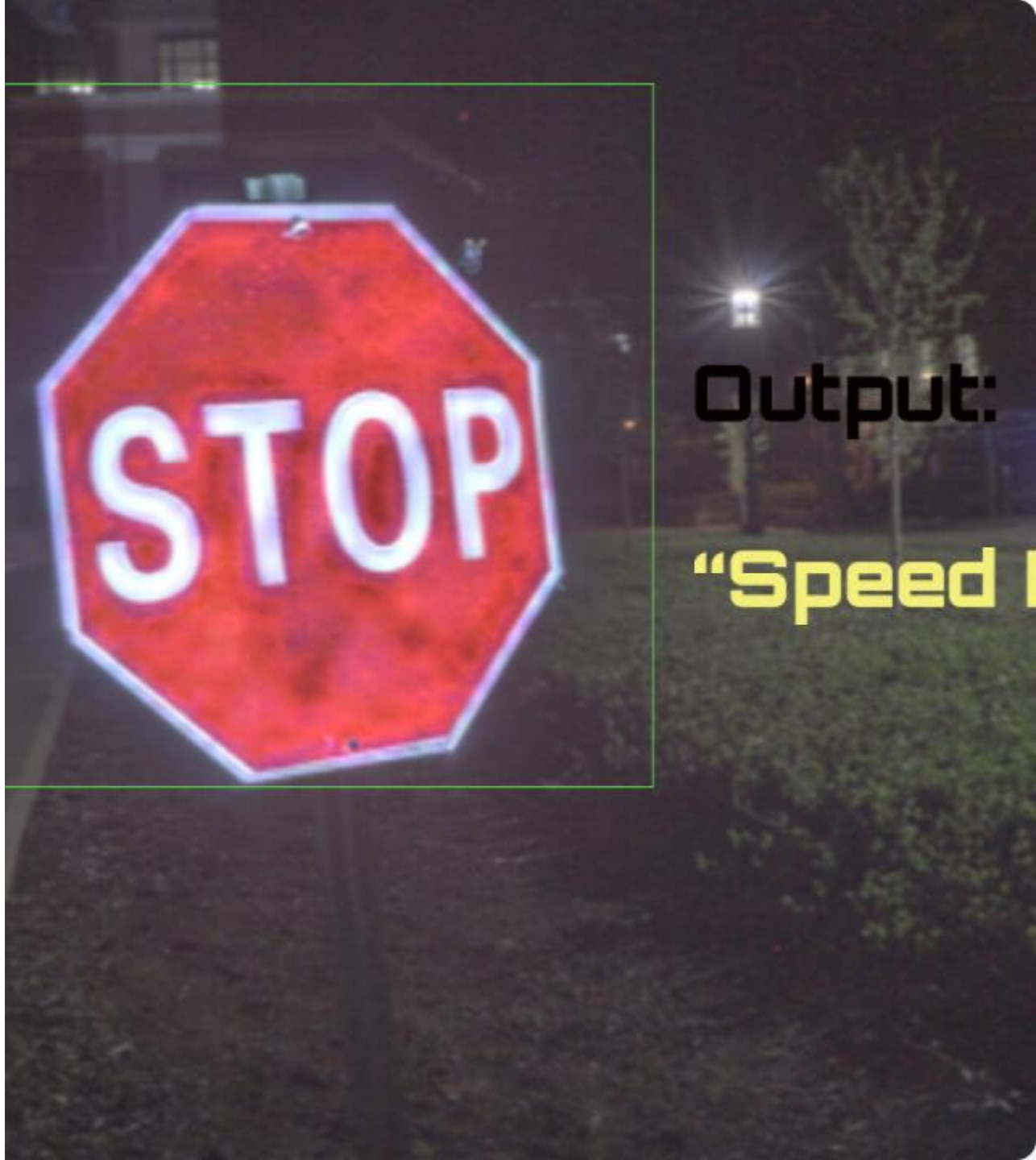
Creators (NYT, Artists) argue that AI is a commercial collage tool that repackages their work without payment, directly competing with them in the market.

Security: Adversarial Machine Learning

Hacking the Model

AI computer vision does not "see" like humans. It sees texture and pixel patterns.

- **The Attack:** Placing a carefully crafted sticker on a Stop sign can trick a self-driving car into identifying it as a "Speed Limit 45" sign.
- **The Threat:** As we deploy AI into physical infrastructure (cars, drones), these "invisible" exploits become life-threatening security risks.

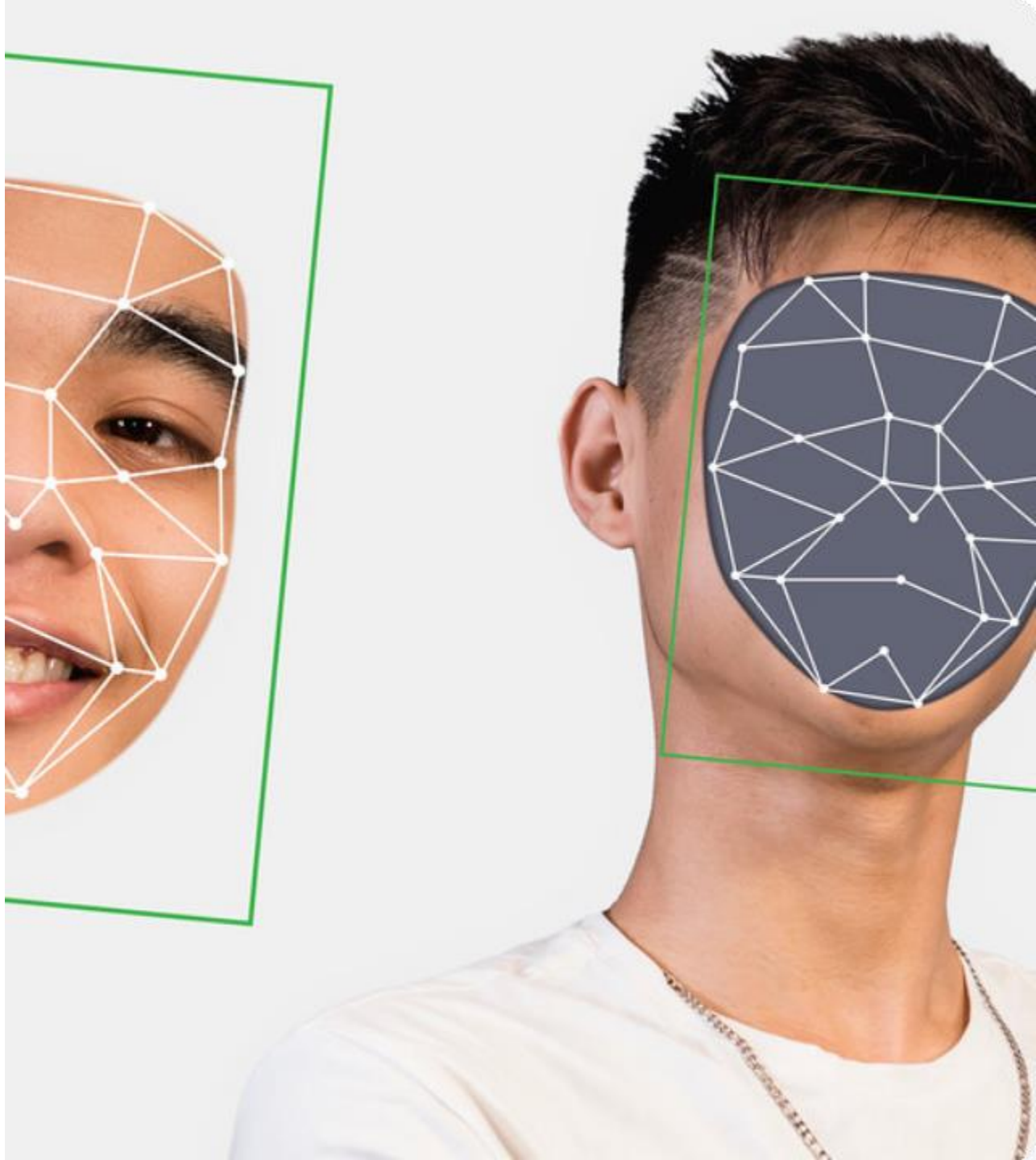


Deepfakes & Authenticity

The Erosion of Reality

Generative Adversarial Networks (GANs) allow for the hyper-realistic swapping of faces and voices. This technology poses severe ethical risks.

- **Misinformation:** Political manipulation through fake videos of leaders.
- **The "Liar's Dividend":** As fake content becomes common, bad actors can dismiss *real* evidence as "just a deepfake," eroding trust in truth itself.
- **Consent:** Using a person's likeness without permission is a fundamental violation of dignity.



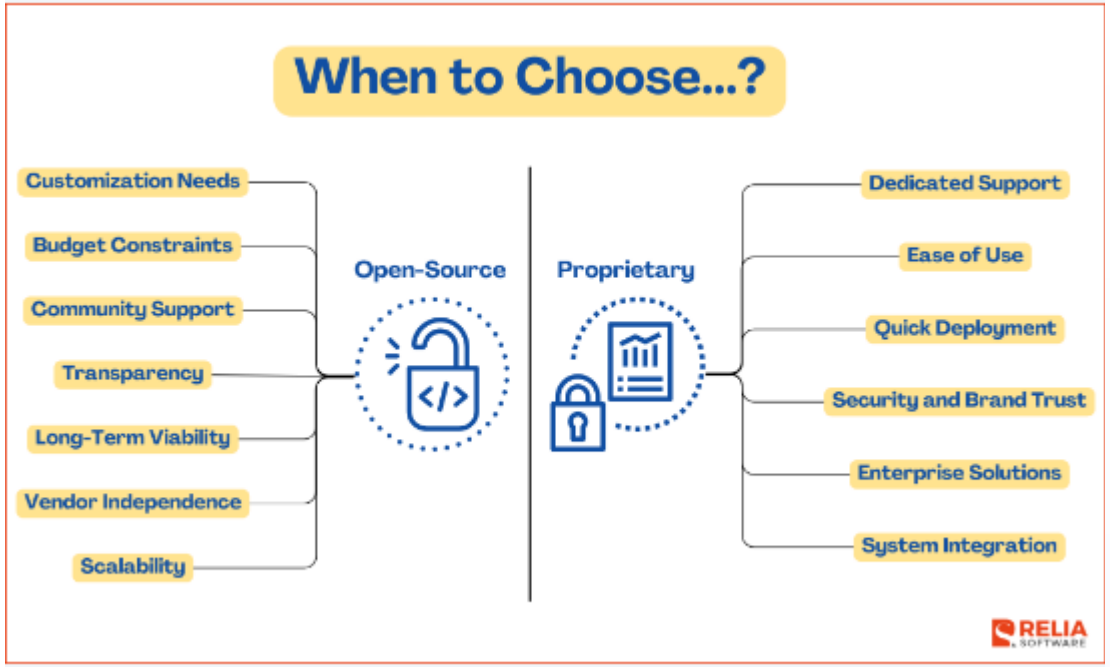
Policy: The Open vs. Closed Debate

Open Weights (e.g., Llama, Mistral)

Pros: Democratizes access, allows researchers to audit for safety/bias, prevents monopoly.
Cons: Bad actors can remove safety guardrails to generate malware/disinfo.

Closed Models (e.g., GPT-4, Gemini)

Pros: Vendor controls safety, can patch vulnerabilities instantly.
Cons: "Security through obscurity," power concentration in few companies.



Discussion: The Liability Gap

The Scenario

A hospital purchases an AI diagnostic tool to help detect lung cancer. The AI analyzes an X-ray and says "Healthy." The doctor, trusting the tool, sends the patient home.

Six months later, the patient is diagnosed with Stage 4 cancer that was visible in the original scan.

1. Is the doctor negligent?

2. Is the software company liable?

3. Does the "Human-in-the-loop" absolve the AI maker?

The Developer's Responsibility

Human-in-the-Loop

Critical systems (medical, judicial) should never be fully automated. A human must always review the AI's recommendation before final action.

Algorithmic Auditing

Testing software not just for bugs, but for bias. Using datasets specifically designed to break the model or expose prejudice.

Transparency

The ethical duty to disclose when a user is interacting with an AI. Deception for the sake of "smooth UX" is a violation of trust.

Theory: Moral Crumple Zones

Who absorbs the impact?

Concept: Just as the crumple zone of a car is designed to absorb the force of impact to protect the passenger, humans in automated systems often absorb the legal and moral liability to protect the system's integrity.

Example: When an autopilot fails, the human driver is blamed for "not paying attention," even if the system was designed to invite complacency.

Regulation: The EU AI Act

The EU has introduced the world's first comprehensive AI law, categorizing tools by risk level:

Risk Level	Description	Examples
Unacceptable	**Prohibited.** Deemed a clear threat to safety, livelihoods, and rights.	Social scoring systems, real-time remote biometric ID in public spaces.
High Risk	**Strict Obligations.** Requires conformity assessments, high-quality data, and logging.	CV-scanning for hiring, medical devices, credit scoring, border control tools.
Limited Risk	**Transparency.** Users must know they are interacting with a machine.	Chatbots (Customer service), Emotion recognition systems, Deepfakes (must be labeled).
Minimal Risk	**No Restrictions.** The vast majority of AI systems currently in use.	Spam filters, AI-enabled video games, Inventory management.

Case Study: Automated Hiring

The Intent

A major tech giant built an AI tool to crawl the web and score job applicants (1 to 5 stars) to automate the search for top talent.

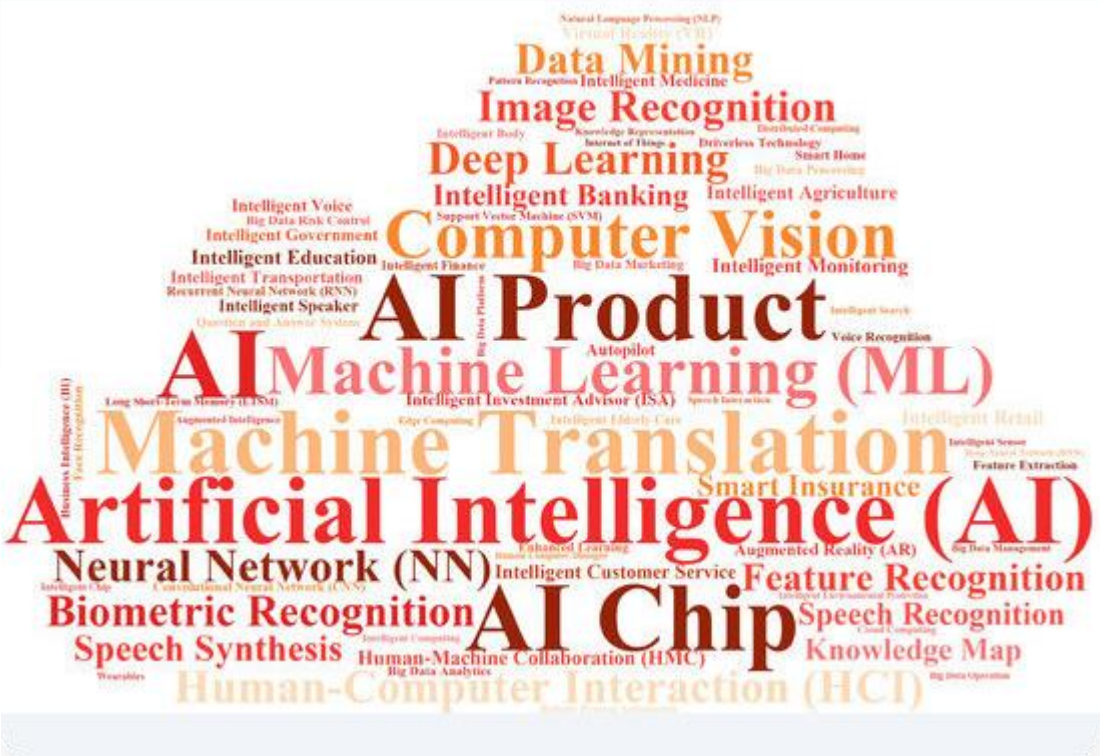
Goal: Remove human subjectivity and speed up recruitment.

The Failure

The model was trained on resumes submitted over a 10-year period (mostly men). The AI learned to penalize resumes containing the word "women's" (e.g., "Women's Chess Club") and downgraded graduates of all-women's colleges.

Result: The tool was scrapped. It proved that AI amplifies existing societal biases if not explicitly corrected.

Context: The Problem of Proxy Variables



You can't just "Remove Gender"

The company removed the "Gender" column from the training data, assuming this would make the AI fair.

- **Proxy Variables:** The AI found other words that *correlated* with gender.
- **Examples:** "Women's College" (strong negative weight), "Jared" or "High School Football" (positive weight).
- **Lesson:** Fairness requires de-biasing the *dataset*, not just hiding the label.

Action: Ethical by Design Checklist

As future professionals, you must integrate ethics into the SDLC (Software Development Life Cycle):

- **Data Diversity:** actively seek out edge cases and under-represented groups in your training data.
- **Model Cards:** Document exactly what the model does, its limitations, and its intended use case (like a nutrition label for code).
- **Red Teaming:** Hire "adversaries" to try and break your model or force it to generate toxic content before release.
- **Continuous Monitoring:** AI models "drift" over time. Monitor for new biases that emerge post-deployment.

Questions?

Data Protection, Ethics & Professional Issues